# LEMMA NORMALIZATION IN SPACES-MORPHISMS.TEX

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*Editorial note.* All mathematical statement and proof text is reproduced verbatim from the author's chapter spaces-morphisms.tex of the Stacks Project, fixed at commit a04446e57ec1fbc252a871afcec7752fb2807b14. The pinned preamble is supplied as its own file; the wrapper class, the xr-hyper wiring and the \externaldocument list are omitted because the assembler supplies the class.

*Editorial note.* Numbering inside this standalone excerpt is editorial, not the full Stacks book print numbering.

*Editorial note.* The candidate is from the corrected band-I follow-on screen (line C) of the extractor's stacks-candidates line: its author label is absent from the live master in every form searched and it belongs to none of the earlier judged pools. The H2 judgement and its reason were made in this module by reading the author's own statement and complete proof in the fixed checkout. Original candidate key: stacks:spaces-morphisms.tex:lemma-normalization.

*Editorial note.* Reference labels in the proof resolve to navigation pointers to the original author source; these pointers do not reproduce proofs of the prerequisites.

*Allowed standard prerequisites (editorial).*

- Prerequisites named by the author inside the counted span. The counted span is reproduced verbatim, including the author's own citations. Its prerequisites are the earlier results of the same fixed Stacks Project version that the author cites there and in the chapter's lead-in.

*Source reference.* Original lemma 0BB1: sources/spaces-morphisms.tex, source lines 10027–10027. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original section 035E: sources/morphisms.tex, source lines 14356–14356. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original definition 035N: sources/morphisms.tex, source lines 14379–14379. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 035Q: sources/morphisms.tex, source lines 14504–14504. Navigation pointer used in the counted span; the span contains the pointer

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*Source reference.* Original lemma 0GMD: sources/spaces-morphisms.tex, source lines 10108–10108. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 07TD: sources/more-morphisms.tex, source lines 5410–5410. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 0262: sources/spaces.tex, source lines 1171–1171. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original theorem 02WW: sources/spaces.tex, source lines 1603–1603. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 07S3: sources/groupoids.tex, source lines 3918–3918. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

**Lemma 0.1.** *Let $S$ be a scheme. For every algebraic space $X$ over $S$ satisfying the equivalent conditions of Lemma 0BB1 there exists a morphism of algebraic spaces*

$$\nu_X : X^\nu \longrightarrow X$$

*with the following properties*

1. *if $X$ satisfies the equivalent conditions of Lemma 0BB1 then $X^\nu$ is normal and $\nu_X$ is integral,*
2. *if $X$ is a scheme such that every quasi-compact open has finitely many irreducible components, then $\nu_X : X^\nu \to X$ is the normalization of $X$ constructed in Morphisms, Section 035E,*
3. *if $f : X \to Y$ is a morphism of algebraic spaces over $S$ which both satisfy the equivalent conditions of Lemma 0BB1 and every codimension $0$ point of $X$ is mapped by $f$ to a codimension $0$ point of $Y$, then there is a unique morphism $f^\nu : X^\nu \to Y^\nu$ of algebraic spaces over $S$ such that $\nu_Y \circ f^\nu = f \circ \nu_X$, and*
4. *if $f : X \to Y$ is an étale or smooth morphism of algebraic spaces and $Y$ satisfies the equivalent conditions of Lemma 0BB1, then the hypotheses of (3) hold and the morphism $f^\nu$ induces an isomorphism $X^\nu \to X \times_Y Y^\nu$.*

*Proof.* Consider the category $\mathcal{C}$ whose objects are the schemes $U$ over $S$ such that every quasi-compact open of $U$ has finitely many irreducible components and whose morphisms are those morphisms $g : U \to V$ of schemes over $S$ such that every generic point of an irreducible component of $U$ is mapped to the generic point of an irreducible component of $V$. We have already shown that

(a) for $U \in \mathrm{Ob}(\mathcal{C})$ we have a normalization morphism $\nu_U : U^\nu \to U$ as in Morphisms, Definition 035N,
(b) for $U \in \mathrm{Ob}(\mathcal{C})$ the morphism $\nu_U$ is integral and $U^\nu$ is a normal scheme, see Morphisms, Lemma 035Q,
(c) for every $g : U \to V \in \mathrm{Arrows}(\mathcal{C})$ there is a unique morphism $g^\nu : U^\nu \to V^\nu$ such that $\nu_V \circ g^\nu = g \circ \nu_U$, see Morphisms, Lemma 035Q part (4) applied to the composition $X^\nu \to X \to Y$,
(d) if $V \in \mathrm{Ob}(\mathcal{C})$ and $g : U \to V$ is étale or smooth, then $U \in \mathrm{Ob}(\mathcal{C})$ and $g \in \mathrm{Arrows}(\mathcal{C})$ and the morphism $g^\nu$ induces an isomorphism $U^\nu \to U \times_V V^\nu$, see Lemma 0GMD and More on Morphisms, Lemma 07TD.

Our task is to extend this construction to the corresponding category of algebraic spaces $X$ over $S$.

Let $X$ be an algebraic space over $S$ satisfying the equivalent conditions of Lemma 0BB1. Let $U \to X$ be a surjective étale morphism where $U$ is a scheme. Set $R = U \times_X U$ with projections $s, t : R \to U$ and $j = (t, s) : R \to U \times_S U$ so that $X = U/R$, see Spaces, Lemma 0262. Observe that $U$ and $R$ are objects of $\mathcal{C}$ by our assumptions on $X$ and that the morphisms $s$ and $t$ are étale morphisms of schemes over $S$. By (a) we have the normalization morphisms $\nu_U : U^\nu \to U$ and $\nu_R : R^\nu \to R$, by (d) we have morphisms $s^\nu : R^\nu \to U^\nu$, $t^\nu : R^\nu \to U^\nu$ which define isomorphisms $R^\nu \to R \times_{s,U} U^\nu$ and $R^\nu \to U^\nu \times_{U,t} R$. It follows that $s^\nu$ and $t^\nu$ are étale (as they are isomorphic to base changes of étale morphisms). The induced morphism $j^\nu = (t^\nu, s^\nu) : R^\nu \to U^\nu \times_S U^\nu$ is a monomorphism as it is equal to the composition

$$\begin{aligned} R^\nu &\to (U^\nu \times_{U,t} R) \times_R (R \times_{s,U} U^\nu) \\ &= U^\nu \times_{U,t} R \times_{s,U} U^\nu \\ &\xrightarrow{j} U^\nu \times_U (U \times_S U) \times_U U^\nu \\ &= U^\nu \times_S U^\nu \end{aligned}$$

The first arrow is the diagonal morphism of $\nu_R$. (This tells us that $R^\nu$ is a subscheme of the restriction of $R$ to $U^\nu$.) A formal computation with fibre products using property (d) shows that $R^\nu \times_{s^\nu, U^\nu, t^\nu} R^\nu$ is the normalization of $R \times_{s,U,t} R$. Hence the étale morphism $c : R \times_{s,U,t} R \to R$ extends uniquely to $c^\nu$ by (d). The morphism $c^\nu$ is compatible with the projection $\mathrm{pr}_{13} : U^\nu \times_S U^\nu \times_S U^\nu \to U^\nu \times_S U^\nu$. Similarly, there are morphisms $i^\nu : R^\nu \to R^\nu$ compatible with the morphism $U^\nu \times_S U^\nu \to U^\nu \times_S U^\nu$ which switches factors and there is a morphism $e^\nu : U^\nu \to R^\nu$ compatible with the diagonal morphism $U^\nu \to U^\nu \times_S U^\nu$. All in all it follows that $j^\nu : R^\nu \to U^\nu \times_S U^\nu$ is an étale equivalence relation. At this point we may and do set $X^\nu = U^\nu/R^\nu$ (Spaces, Theorem 02WW). Then we see that we have $U^\nu = X^\nu \times_X U$ by Groupoids, Lemma 07S3.

What have we shown in the previous paragraph is this: for every algebraic space $X$ over $S$ satisfying the equivalent conditions of Lemma 0BB1 if we choose a surjective étale morphism $g : U \to X$ where $U$ is a scheme, then we obtain a cartesian diagram

$$\begin{array}{ccc} X^\nu & \xleftarrow{g^\nu} & U^\nu \\ {\scriptstyle \nu_X}\downarrow & & \downarrow{\scriptstyle \nu_U} \\ X & \xleftarrow{g} & U \end{array}$$

of algebraic spaces. This immediately implies that $X^\nu$ is a normal algebraic space and that $\nu_X$ is an integral morphism. This gives part (1) of the lemma.

We will show below that the morphism $\nu_X : X^\nu \to X$ up to unique isomorphism is independent of the choice of $g$, but for now, if $X$ is a scheme, we choose $\mathrm{id} : X \to X$ so that it is clear that we have part (2) of the lemma.

We still have to prove parts (3) and (4). Let $g : U \to X$ and $\nu_X : X^\nu \to X$ and $g^\nu : U^\nu \to X^\nu$ be as above. Let $Z$ be a normal scheme and let $h : Z \to U$ and $a : Z \to X^\nu$ be morphisms over $S$ such that $g \circ h = \nu_X \circ a$ and such that every irreducible component of $Z$ dominates an irreducible component of $U$ (via $h$). By Morphisms, Lemma 035Q part (4) we obtain a unique morphism $h^\nu : Z \to U^\nu$ such that $h = \nu_U \circ h^\nu$. Picture:

$$\begin{array}{ccccc} & & & \overset{a}{\curvearrowleft} & \\ X^\nu & \xleftarrow{g^\nu} & U^\nu & \xleftarrow{h^\nu} & Z \\ \downarrow{\scriptstyle \nu_X} & & \downarrow{\scriptstyle \nu_U} & \swarrow{\scriptstyle h} & \\ X & \xleftarrow{g} & U & & \end{array}$$

Observe that $a = g^\nu \circ h^\nu$. Namely, since the square with corners $X^\nu$, $X$, $U^\nu$, $U$ is cartesian, this follows immediately from the fact that $h^\nu$ is unique (given $h$). In other words, given $h : Z \to U$ as above (and not $a$) there is a unique morphism $a : Z \to X^\nu$ with $\nu_X \circ a = g \circ h$.

Let $f : X \to Y$ be as in part (3) of the statement of the lemma. Suppose we have chosen surjective étale morphisms $U \to X$ and $V \to Y$ where $U$ and $V$ are schemes such that $f$ lifts to a morphism $g : U \to V$. Then $g \in \mathrm{Arrows}(\mathcal{C})$ and we obtain a unique morphism $g^\nu : U^\nu \to V^\nu$ compatible with $\nu_U$ and $\nu_V$. However, then the two morphisms

$$R^\nu = U^\nu \times_{X^\nu} U^\nu \to U^\nu \to V^\nu \to Y^\nu$$

must be the same by our comments in the previous paragraph (applied with $Y$ in stead of $X$). Since $X^\nu$ is constructed by taking the quotient of $U^\nu$ by $R^\nu$ it follows that we obtain a (unique) morphism $f^\nu : X^\nu \to Y^\nu$ as stated in (3).

To see that the construction of $X^\nu$ is independent of the choice of $g : U \to X$ surjective étale, apply the construction in the previous paragraph to $\mathrm{id} : X \to X$ and a morphism $U' \to U$ between étale coverings of $X$. This is enough because given any two étale coverings of $X$ there is a third one which dominates both. The reader shows that the morphism between the two normalizations constructed using either $U' \to X$ or $U \to X$ becomes an isomorphism after base change to $U'$ and hence was an isomorphism. We omit the details.

We omit the proof of (4) which is similar; hint use part (d) above. $\square$